

Scene-Assimilated Video World Models

Retrieval-guided data assimilation for image-to-video generation

Core idea. A retrieved real video is not just extra prompt context. It is treated as a sparse scene observation that corrects an open-loop video forecast.

Forecast

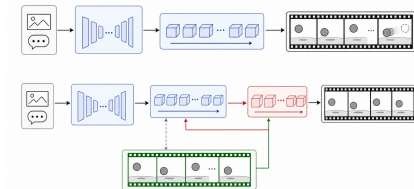
image-to-video rollout from anchor image and prompt

Observation

retrieved real-video trajectory

Analysis

denoising representation after observation fusion



Motivation: Open-Loop Video World Models

PROBLEM

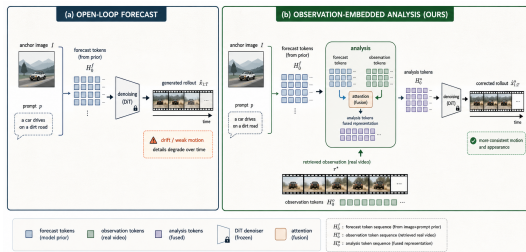


Image-to-video generation behaves like a world model.

- ▶ Input: one anchor image and a short prompt.
- ▶ Output: a plausible future trajectory of the scene.
- ▶ Failure point: the model must infer dynamics that text and one image do not specify.

Prompts such as “a surfer riding a wave” omit contact timing, acceleration, camera motion, water deformation, and scene-level parallax.

Thesis: Retrieval as Sparse Scene Observation

PARADIGM

Reframing retrieval

- ▶ Retrieval is often used as semantic context or a reference to copy.
- ▶ Here, the retrieved clip is a partial empirical trajectory from a related real scene.
- ▶ The clip should correct under-specified dynamics without overwriting the anchor image.

Data assimilation view

$$p(x_t \mid y_{1:t}) \propto p(y_t \mid x_t) p(x_t \mid y_{1:t-1})$$

$$x^a = x^f + \frac{P^f}{P^f + R}(y - x^f)$$

The equations are not claimed as exact for video DiTs; they define the role of each component: forecast, observation, innovation, and analysis.

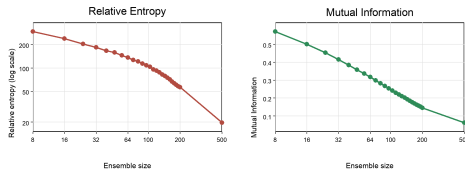
Scene-assimilated generation asks where real observations enter the latent transition state of the generator.

Controlled Evidence: Moving MNIST Latent ETKF

SANITY CHECK

Why this experiment?

- ▶ The final video generator is too expensive for explicit ensemble DA.
- ▶ Moving MNIST gives a controlled setting where latent state correction can be measured.
- ▶ If observation is genuinely assimilated, the information gap between forecast and analysis should shrink.



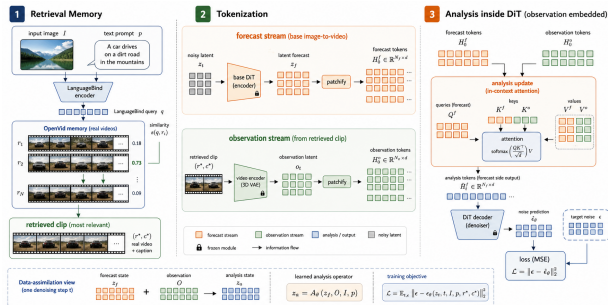
ETKF ensemble	$D_{\text{KL}} \downarrow$	residual $I(Z; Y) \downarrow$
8 members	295.4926	0.5709
100 members	105.9172	0.2477
500 members	19.6006	0.0622

$$D_{\text{KL}}(p_a \| p_f) = \int p_a(z) \log \frac{p_a(z)}{p_f(z)} dz$$

$$I(Z; Y) = H(Z) - H(Z | Y)$$

Method Overview

RETRIEVAL-GUIDED DATA ASSIMILATION



Three components

- **Retrieval memory:** OpenVid-style real videos indexed by LanguageBind.
- **Two streams:** forecast tokens from the generator and observation tokens from retrieved video.
- **Analysis update:** in-context DiT attention fuses both streams before noise prediction.

Retrieval Memory as Observation Source

OBSERVATION SELECTION

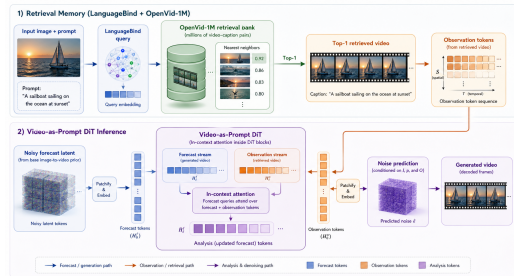
Memory bank

$$\mathcal{R} = \{(r_i, c_i)\}_{i=1}^N$$

$$q = Q_{LB}(I, p)$$

$$(r^*, c^*) = \arg \max_{(r_i, c_i) \in \mathcal{R}} s(q, r_i)$$

The query combines anchor image I and prompt p . The retrieved clip r^* is a nearby real-world trajectory, not the target video.



Analysis Inside the DiT

LEARNED STATE UPDATE

Forecast and observation streams

$$H_0^f = P_{\text{patch}}(z_t^f, l, p)$$
$$O = E_{\text{vid}}(r^*, c^*), \quad H_0^o = P_{\text{obs}}(O)$$

The forecast stream preserves the generator's own rollout. The observation stream carries real-video dynamics.

Attention as analysis operator

$$\bar{H}_\ell^f = \text{Attn} \left(Q_\ell^f, \begin{bmatrix} K_\ell^f \\ K_\ell^o \end{bmatrix}, \begin{bmatrix} V_\ell^f \\ V_\ell^o \end{bmatrix} \right)$$

$$z_t^a = \mathcal{A}_\theta(z_t^f, O, l, p)$$

$$\hat{e} = \epsilon_\theta(z_t^f, t, l, p, O)$$

Fine-Tuning Objective

TRAINING

What is trained?

- ▶ Keep the base image-to-video DiT, VAE, and text encoder frozen.
- ▶ Fine-tune the retrieval-guided in-context pathway.
- ▶ Use the target clip for denoising supervision and a non-identical retrieved clip as observation.

Conditional diffusion loss

$$\mathcal{L}_{\text{ft}} = \mathbb{E}_{x, l, p, r^*, c^*, t, \epsilon} \left[\|\epsilon - \epsilon_{\theta}(z_t, t, l, p, r^*, c^*)\|_2^2 \right]$$

This keeps the architecture modular: base generator as forecast model, retrieval bank as world memory, in-context pathway as analysis update.

Wan2.1 Pilot Setup

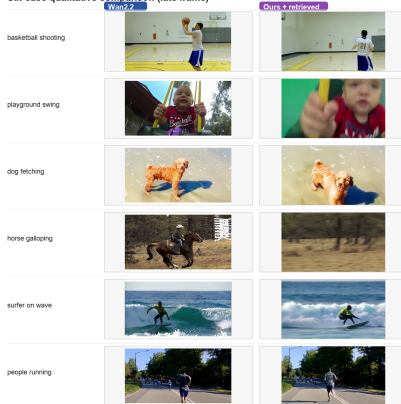
QUALITATIVE EVALUATION

Comparison

- ▶ Baseline: Wan2.1-14B image-to-video from image and prompt.
- ▶ Ours: same input plus one retrieved real-video observation through the VAP-style pathway.
- ▶ Six short-prompt cases: basketball, swing, dog, horse, surfing, and running.

A mechanism check under prompt under-specification, not a completed benchmark.

Six-case qualitative comparison (late frame)

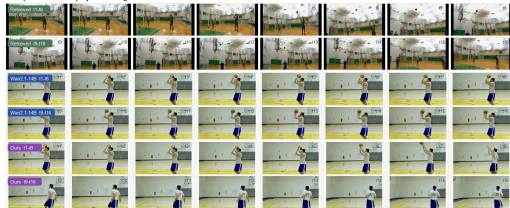


Qualitative Results: Motion Becomes More Active

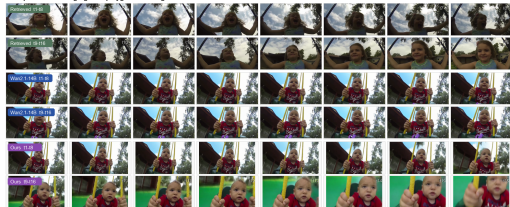
RESULTS

Qualitative temporal samples: cases 1-2

Prompt: a basketball player shooting



Prompt: a child swinging on a playground swing



Observed pattern

- ▶ Basketball: clearer shooting trajectory and follow-through.
- ▶ Swing: stronger depth and camera motion.
- ▶ Dog, horse, surfing, running: more body, scene, and background dynamics.

Retrieval helps most when the prompt leaves physical evolution ambiguous and the retrieved clip is spatially compatible.

Failure Modes Are Part of the Claim

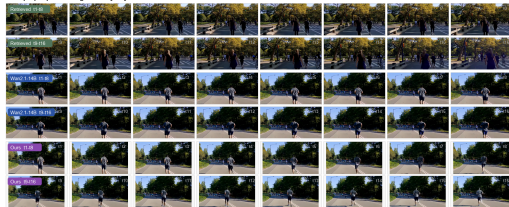
LIMITS

Qualitative temporal samples: cases 5-6

Prompt: a surfer riding a small wave



Prompt: people running on a city skywalk



Observation can be too strong

- ▶ Mismatched viewpoint can introduce framing drift.
- ▶ Mismatched subject scale or identity can soften details.
- ▶ More motion is not automatically a better world model.

The real tradeoff is physical consistency versus spatial anchoring. A robust analysis operator must learn how much to trust the retrieved observation.

Benchmark Implication

WHAT SHOULD BE MEASURED

Motion

Dynamic degree, action accuracy, temporal smoothness, optical-flow plausibility.

Spatial anchoring

Subject consistency, background consistency, identity preservation, framing stability.

World-model behavior

Physical consistency, causal event evolution, retrieval-alignment stratification.

Expected result is conditional, not uniform.

- ▶ Aligned observations should improve motion and physical consistency.
- ▶ Mismatched observations may reduce identity, sharpness, or background stability.
- ▶ Reporting only aggregate quality can hide whether the model actually assimilates useful evidence.

Takeaways

CONCLUSION

Contributions

- ▶ Formulate image-to-video generation as open-loop latent forecasting.
- ▶ Introduce retrieved real videos as sparse scene observations.
- ▶ Implement scene-assimilated analysis with LanguageBind retrieval and a VAP-style in-context DiT pathway.
- ▶ Provide Moving MNIST and Wan2.1 pilot evidence for the promise and limits of the idea.

Future work

- ▶ Retrieval filtering by viewpoint, identity, background, and camera motion.
- ▶ Observation confidence or uncertainty-aware fusion.
- ▶ Full benchmark with VBench, FVD/FID, and physical-consistency tests.
- ▶ More explicit 3D scene state for geometry-aware assimilation.

Retrieval should be evaluated as an estimator: does aligned evidence close the missing-dynamics gap without overwriting anchor-state variables?

Thank you

Questions?

Scene-assimilated video world models treat retrieval as observation, attention as analysis, and generation as corrected latent forecasting.